CHAPTER 1

INTRODUCTION

Growing concerns about environmental pollution and the overuse of synthetic plastics have intensified the demand for sustainable alternatives. Starch-based edible films have emerged as a viable solution due to their biodegradability, edibility, renewability, and affordability. Among various starch sources, tapioca starch, derived from cassava roots (*Manihot esculenta*), is particularly suitable for film formation owing to its high amylopectin content, neutral taste, and excellent film-forming properties.

The quality of starch films is significantly influenced by the interaction between starch, water, and plasticizers such as glycerol. These interactions affect key properties like mechanical strength, moisture resistance, and flexibility. In this study, edible films were formulated using tapioca starch as the base polymer, glycerol as the plasticizer, and water as the dispersing agent.To achieve optimal film properties, Response Surface Methodology (RSM) was used to systematically evaluate and optimize the concentrations of each component. This statistical approach reduced experimental effort while identifying ideal combinations for improved film performance.

The optimized films were evaluated through tests measuring pencil hardness, tear resistance, water absorption, and compositional attributes like carbohydrate and ash content. Results indicated that the optimized films offered enhanced mechanical strength, better

moisture control, and sufficient durability for use in food packaging applications. This study demonstrates the potential of tapioca starch-based edible films as eco-friendly packaging alternatives. It provides valuable insights into formulation optimization and paves the way for further development and potential commercialization of sustainable packaging materials.

1.1 OBJECTIVES

- To extract the Starch from the Tapioca by three different methods (acid/alkali/sedimentation)
- To compare the film forming properties of the extracted starch from different methods
- To optimize the composition of edible film from the selected starch
- To study the different film properties of the optimized edible film

CHAPTER 2

REVIEW OF LITERATURE

Aghazadeh et al. (2018) investigated the physicochemical properties of cereal starch-based edible films incorporating glycerol as a plasticizer. Their study revealed that glycerol enhanced water vapor barrier properties, reduced tensile strength, and increased elongation. These findings underline glycerol's role in improving the flexibility and barrier properties of starch-based films for packaging applications.

Basiak et al. (2018) explored how varying glycerol and water contents influence the structural and functional properties of starch-based edible films. The study demonstrated that glycerol primarily enhanced film transfer properties and moisture sorption while having minimal impact on color and mechanical properties, underscoring its importance in optimizing functional characteristics for food coatings.

Bergo et al. (2008) examined the influence of plasticizers on the physical properties of cassava starch-based edible films. Their findings suggested that increasing glycerol content improved flexibility but reduced tensile strength and barrier properties, essential considerations for optimizing films intended for food packaging.

Chiumarelli et al. (2014) developed edible films and coatings using cassava starch, glycerol, carnauba wax, and stearic acid. They found that the combination of these ingredients improved the mechanical properties, moisture resistance, and barrier efficiency of the films, making them suitable for food preservation.

Chillo et al. (2008) studied the effects of glycerol and chitosan on the properties of tapioca starch-based edible films. The research demonstrated that glycerol increased film flexibility, while chitosan contributed to improved water vapor permeability and mechanical strength, contributing to the development of better biodegradable films for food packaging.

Cisse et al. (2012) examined the effects of varying chitosan and glycerol concentrations on the properties of edible films. Their study found that higher glycerol concentrations increased flexibility and elongation but decreased tensile strength, supporting the importance of glycerol in optimizing film properties for food packaging.

Farahnaky et al. (2013) explored the impact of glycerol on the physical and mechanical properties of wheat starch-based films. They found that increasing glycerol content enhanced solubility, lightness, and moisture absorption while decreasing tensile strength, emphasizing glycerol's role in improving the flexibility and functional properties of starch-based films for food packaging.

Fama et al. (2005) examined the mechanical properties of tapioca starch-based edible films containing sorbates. They found that sorbates improved flexibility and reduced brittleness, demonstrating how incorporating functional ingredients can enhance film performance, similar to glycerol's role.

Gutierrez et al. (2015) analyzed the structural and mechanical properties of edible films produced from native and modified starches of cush-cush yam and cassava. They found that starch modification improved film strength, flexibility, and stability, advancing the use of underutilized starch sources in sustainable food packaging.

Juliyarsi et al. (2011) evaluated the quality of edible films using glycerol as a plasticizer. They found that increasing glycerol concentration improved film flexibility and elongation but reduced tensile strength. The films also became more transparent and slightly more permeable to moisture, making glycerol a useful additive for enhancing flexibility.

Kaveh et al. (2020) examined the effects of alcoholic-alkaline treatments on the physical and mucoadhesive properties of tapioca starch. Their study showed that these treatments enhanced the starch's adhesive properties and modified its structure, improving its suitability for biodegradable film production.

Kumar (2019) reviewed polysaccharide-based films, highlighting starch for its low cost and good film-forming ability. The study found that acid and alkaline hydrolysis improve starch properties like solubility and flexibility, while glycerol enhances flexibility but may reduce strength and barrier properties.

Kusumaningtyas et al. (2018) studied the preparation and characterization of edible films made from sorghum starch, incorporating glycerol and sorbitol as plasticizers. They found that both glycerol and sorbitol improved film flexibility and mechanical properties while enhancing water vapor permeability, which is crucial for food packaging.

Maran et al. (2013) developed a mathematical model to predict and optimize the mechanical properties of edible films from tapioca starch. They used response surface methodology to examine variables such as plasticizer concentration and drying temperature, focusing on tensile strength and elongation for optimizing biodegradable and eco-friendly packaging materials.

Parra et al. (2004) explored the mechanical behavior and water vapor transmission rates of cassava starch-based edible film blends. The study showed that blending cassava starch with other polymers affected the films' mechanical and barrier properties, supporting their potential use in moisture-sensitive packaging applications.

Pansaeng (2019) investigated the impact of glycerol concentration on the properties of biodegradable films made from cassava starch (Saai Dieow cultivar). The study found that glycerol concentrations influenced film flexibility, thickness, and tensile strength, making it suitable for applications like plant packaging.

Piyawatakarn et al. (2015) studied the effect of glycerol on tapioca starch-based films. The research found that glycerol, as a plasticizer, improved film flexibility and elongation while reducing tensile strength, providing valuable insights for tailoring film properties in biodegradable packaging materials.

Prabha & Ranganathan (2018) focused on optimizing the barrier properties of biodegradable films made from tapioca starch. They identified factors such as film thickness and drying conditions that significantly impacted moisture and oxygen barrier properties, enhancing the performance of tapioca starch films in sustainable packaging.

Rumyaan et al. (2022) investigated the characteristics of edible films made from bitter cassava starch with glycerol. They found that glycerol improved film flexibility, solubility, and moisture permeability, highlighting the potential of cassava starch-based films for sustainable food packaging.

Saberi et al. (2016) examined the properties of pea starch edible films with varying glycerol concentrations. They observed that glycerol enhanced flexibility and elongation while reducing tensile strength, reinforcing its role as an effective plasticizer in balancing film flexibility and strength for packaging applications.

Sartika et al. (2022) reviewed the tensile properties of edible films made from various starches, focusing on glycerol's role as a plasticizer. The study highlighted how glycerol affected mechanical properties such as tensile strength and elongation, providing insights into optimizing film formulations for improved flexibility and durability in food packaging.

Wilfer et al. (2021) studied the effect of different native starch types on edible film properties. They found that starch source significantly influenced film characteristics, emphasizing the importance of starch selection or modification to optimize film performance for biodegradable packaging.

Zhong and Li (2014) examined the effects of glycerol and storage relative humidity on kudzu starch-based edible films. Their study showed that glycerol improved film flexibility and tensile strength, while high humidity reduced the films' mechanical properties, highlighting the need to control storage conditions for maintaining starch-based film quality

CHAPTER3

MATERIALS AND METHODOLOGY

3.1 MATERIALS

3.1.1 Raw materials used

Edible films are made from tapioca starch, glycerol, and water, which each have an effect on the properties of the film. Tapioca starch, which contains a high amylopectin content, is the main film-forming agent, producing smooth, flexible, and biodegradable films that are ideal for green packaging. Glycerol is used as a plasticizer to increase flexibility and elasticity by weakening hydrogen bonding among the starch molecules, producing smooth, shiny, and less brittle films, which are perfect for food contact. Water acts as a dispersion medium and solvent, facilitating starch gelatinization and mixing. Starch contributes structure, glycerol flexibility, and water proper blending to create strong edible films with good properties.

3.2 METHODOLOGY FOR DEVELOPING THEEDIBLE FILM

3.2.1 SEDIMENTATION METHOD OF STARCH EXTRACTION

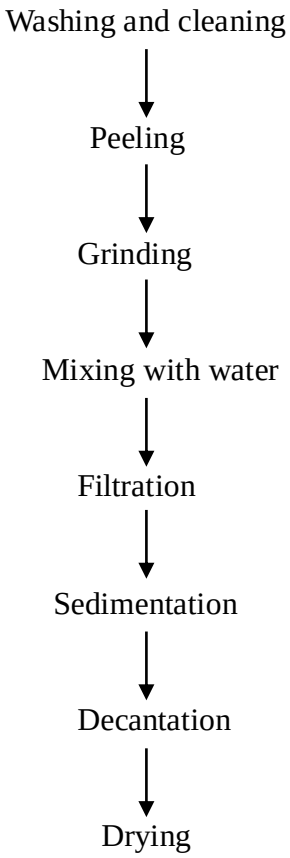


Fig 3.1 Flowsheet of sedimentation method of starch extraction

3.2.2 ALKALINE HYDROLYSIS METHOD OF STARCH EXTRACTION

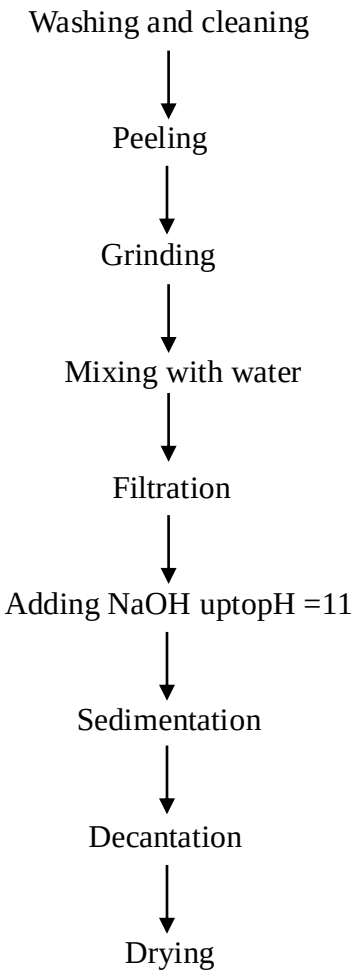


Fig 3.2 Flowsheet of alkaline hydrolysis method of starch extraction

3.2.3 ACID HYDROLYSIS METHOD OF STARCH EXTRACTION

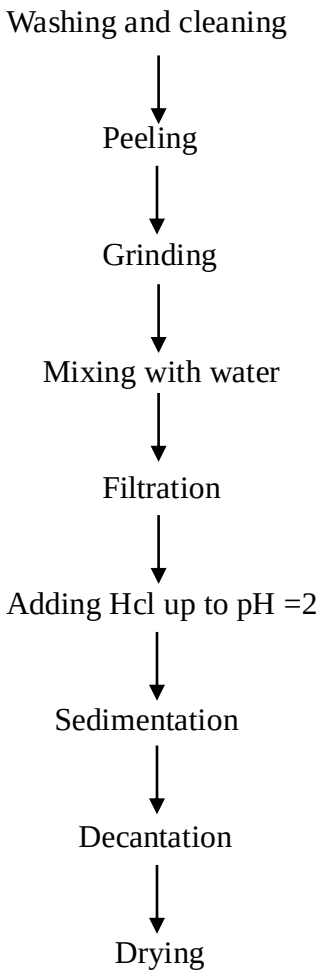


Fig 3.3 Flowsheet of acid hydrolysis method of starch extraction

3.3 ESTIMATION OF ASH CONTENT

The ash content of the developed animal feed has been determined by weighing accurately 10 g of formulated feed into a weighed crucible. It is incinerated in the muffle. The Petri dish was kept in a muffle furnace maintained at 550-600℃ until light grey ash results or to a constant weight then cooled in a desiccator and weighed. The ash was found out from the Equation (3.2).

$$Total\ Ash\ Content = (W_2\text{-}W) \div (W_1\text{-}W) \times 100 \qquad (3.2)$$

Where:

W_1 = Weight of the crucible with the dried material taken for the test in (g)

W_2 = Weight of the crucible with the ash (g)

W = Weight of the empty crucible (g)

3.4ESTIMATION OF CARBOHYDRATES

To extract sugars from starch, homogenize the sample in hot 80% ethanol centrifuge, and wash it multiple times until it no longer shows color. Dry the residue in a water bath, and then add distilled water and 52% perchloric acid to the residue. Centrifuge for 20 min at 0℃, repeating with fresh perchloric acid. Combine the extracts, pipette 0.1 ml into tubes, fill 1.0 ml with pure water, add 4.0 ml of anthrone reagent, and boil in a water bath for 8 min. Cool and measure the color at 620 nm.

3.5 ESTIMATION OF WATER ACTIVITY

Water activity is the measure of amount of free water present in the food for microbial contamination. The water activity of developed feed was measured using Aqualab digital water activity meter.

3.6RESPONSE SURFACE METHODOLOGY (RSM) FOR THE OPTIMIZATION

The edible film was produced through a mixture design approach with optimization via a quadratic model. The basic ingredients in the film consisted of starch, glycerol, and water-all essential factors concerning the film characteristics. The starches, prepared from tapioca, were used in forming the film, in the range of 1.8 g representing the low end to 3 g on the upper end. Glycerol, applied as a plasticizer to confer flexibility and to prevent brittleness, came in between 12 to 18 grams. This water acted as a solvent and dispersion medium that was necessary for starch gelatinization, applied between 40 g and 46.2 g which is given in the Table 3.1. The mixture design allowed the component ratios to be varied systematically, with the quadratic model describing the combined effects of these ratios on film characteristics, such as optimization for flexibility, tensile strength, and moisture barrier properties. Through this, a rugged, biodegradable edible film of quality was assured.

Table 3.1 Upper and lower range of the mixture design

	Name	Low	High
A[Mixture]	Starch	1.8g	3g
B[Mixture]	Glycerol	12g	18g
C[Mixture]	Water	40g	46.2g

3.7 ESTIMATION OF TENSILE STRENGTH

To determine the tensile strength of a single sample of packaging material, a strip measuring 20 mm in width and approximately 125 mm in length is first cut from the film. The testing machine is prepared by rotating the top knob to open the grips. One end of the test sample is securely fixed into the bottom grip and tightened. The top grip is then opened, and the other end of the sample is inserted and clamped. The top grip, attached to the load cell hook, ensures the sample is held in place vertically. Once the sample is properly fixed, the machine is started by pressing the ON button. As the machine operates, it pulls the sample apart while the increasing load and deflection values are displayed on the control panel. The load continues to rise until the sample tears, at which point the stop button is pressed to end the test. Afterward, the grips are returned to their original position by pressing the UP button, and the torn sample is carefully removed. The same procedure can be followed to evaluate the seal strength of the packaging film and this was found from the Equation (3.3).

Tensile strength = average breaking force / width *(3.3)*

3.8 ESTIMATION OF TEAR RESISITANCE

To perform a tear strength test on a single test specimen, the instrument is first placed on a level bench. Before testing, the levelling screw is adjusted to ensure the instrument is properly aligned. With the pendulum in its initial position, the pendulum stop is set so that the index mark on the right-hand corner of the pendulum aligns exactly with the corresponding index mark on the instrument base. The test specimen, typically a rectangular strip of paper or film with a 62 mm edge, is then clamped vertically and centrally in the jaws, ensuring that the lower edge of the sheet rests on the bottom of the jaws and the side edges are properly aligned. A slit is introduced in the specimen by gently depressing the knife handle as far as it will travel, then releasing it. The pendulum stop is then depressed, and as the pendulum swings and begins its return, it is gently caught to prevent overshooting. The reading indicated by the pointer is noted to the nearest 0.1 unit. The pendulum is then returned to its initial position and locked in place by allowing the pendulum stop to rise. The pointer is gently reset against the stop, and the torn specimen is removed. This procedure is repeated for the remaining specimens, ensuring that at least six tests are conducted in each principal direction of the paper to obtain consistent and reliable resultsfrom Equation (3.4).

Tearing strength = (tear resistance – dead weight) / no.of poles *(3.4)*

3.9 ESTIMATION OF WATER VAPOUR PERMEABILITY OF EDIBLE FILM

For the purpose of studying the water vapor permeability of a particular packaging material, a certain volume of water is first measured out and placed in a 250 ml glass vial. This vial containing the water is weighed on a sensitive weighing balance to obtain the initial weight. The next step involves placing over the mouth of the vial the packaging film sample that covers the mouth entirely and securing it tightly with a rubber band to provide a sealed fit. The preparation is kept undisturbed for one day under controlled environmental conditions.

After the exposure time, the packaging material is then carefully removed from the vial, and the remaining water in the vial is weighed separately with the packaging sample. If the addition of the weight of the vial and water does not change from the previously recorded value, then it indicates that there has not been any loss of water vapor through the packaging, inferring that the said material is water vapor impermeable. This simple but effective test enables the assessment of the moisture barrier properties of the packaging film from the Equation (3.5).

Permeability of sample = (difference in weight of beaker + water + sample) /
Area of beaker mouth *(3.5)*

3.10 ESTIMATION OF WATER ABSORPTION CAPACITY OF EDIBLE FILM

To conduct the COBB test for measuring water absorption of paper and paperboard, the dry sample should first be weighed on a digital balance with 0.01 g accuracy. This figure shall be recorded as initial weight. Proceed to pour exactly 100 ml of clean water- into the testing bowl, using the measuring jar that comes with the COBB tester. The paper sample should be put over the bowl so that the glazed side or treated side of the paper faces the water surface. The top lid would be placed over the paper, securing and clamping it in place by tightening the top knob.

When secured, the whole assembly would be inverted with the help of the operating handle so as to allow water exposure to the paper. For the 60-second COBB test, keep this position steady for 50 seconds; through the120-second test, hold for 110 seconds. This pre-timed allowance ensures that by the time you put the apparatus back to its position, the exposure duration coincides with the test standard.

After the elapsed time, turn back the assembly to its position using the handle. Unscrewing the top knob, remove the lid carefully. Subsequently, take out the wet sample and put it between two layers of blotting paper to remove the surplus of surface water. He rolled it once forward and once backward using a 10 kg roller. It must be noted that the rolling is done without any extra pressure applied apart from the roller weight and the final value is calculated from the Equation(3.6).

$$\textit{Water absorption capacity}(g/m^2) = \textit{difference in weight / area} \qquad (3.6)$$

3.11 ESTIMATION OF BIODEGRADABILITY OF THE EDIBLE FILM

The biodegradability of the edible film made from tapioca starch, glycerol, and water was assessed through a soil burial test. The film samples were buried in natural soil at a specific depth and monitored for degradation. Visual inspection and weight loss measurements were used to evaluate biodegradation. The films exhibited noticeable disintegration over time, indicating that the film is highly biodegradable and suitable for eco-friendly applications.

CHAPTER4

RESULT AND DISCUSSION

The study was able to show that edible films can be made with tapioca starch, glycerol, and water. The films showed promising results in strength, barrier property, and sustainability for food packaging. Being a high amylopectin starch, tapioca starch formed the structural matrix, whereas glycerol acted as a plasticizer, which has great flexibility and less brittleness. Water was a solvent that aided in starch gelatinization and uniform film formation. Starch extracted under alkaline hydrolysis produces cleaner starch, which improved the texture and strength of the films. Films are stable in moisture and ash content, while the carbohydrate content remains high, thus confirming the essential role of starch.

Mechanical testing showed adequate tensile and tear strength, with glycerol contributing to elasticity. In terms of barrier properties, they exhibited moderate water vapor permeability; films do present hydrophilic properties, but they can be controlled with appropriate levels of glycerol. By using the Response Surface Methodology, an optimum formulation was obtained: starch (1.8-3 g), glycerol (12-18 g), and water (40-46.2 g), thereby providing good flexibility, strength, and moisture control. In conclusion, the optimized films are biodegradable, transparent, and suitable for various eco-friendly food packaging applications.

4.1 OPTIMIZATION OF STARCH EXTRACTION METHOD

Table4.1 optimization of starch extraction method

Parameters	Sedimentation method	Acid hydrolysis method	Alkaline hydrolysis method
Gelatinization temperature	66℃	-	63℃
Gelatinization time	5 min 21 sec	-	3 min 54 sec
Viscosity	190.8 N/m^2	117.7N/m^2	555.6N/m^2
Yield	39.8%	21.7%	26.7%

In comparison with several starch extraction methods, namely Sedimentation, Acid Hydrolysis, and Alkaline Hydrolysis, the Sedimentation Method has been identified as the best option due to the balanced performance it offers in the gelatinization temperature, time, viscosity, and yield. The gelatinization temperature makes, tapioca flour gelatinizes most efficiently at about 65°C, allowing it to maintain structural integrity during processing. The sedimentation method is close to the ideal temperature of 66°C and hence can be regarded as an effective method. The gelatinization time Between 5 and 15 minutes are required for the viscosity and gel formation to reach maximum gelatinization. With a time of 5 minutes and 21 seconds, the sedimentation method fell within this range and thus is considered efficient.

The viscosity values suitable for good starch films are usually between 113 and 255 cp. The sedimentation method presents a viscosity of 190.8 cp, suggestive of a good potential for film formation and starch integrity. High yield ensures commercial viability. Sedimentation, with a yield of 39.8%; acid hydrolysis, 21.7%; and alkali hydrolysis, 26.7%,

displayed the best efficiency. In conclusion, the optimal method for commercial or research-scale starch extraction is sedimentation, given its moderate processing conditions, high yield associates, and the literature.

4.2 PRELIMINARY TRIALS

Table4.2 Preliminary trials composition of edible film

No of trials	Raw materials			Response		
	Starch(g)	Glycerol(g)	Water(g)	Gelatinization Temperature	Gelatinization Time	Viscosity
Trial-1	1.8	12	46.2	65℃	5min 9sec	120.4cp
Trial-2	2.4	15	42.6	68℃	5min 56sec	148.7cp
Trial-3	3	18	39	70℃	6min 51sec	174.2cp

In the first stages of the investigation, three trials were performed to assess the effects of different starch amounts on the thermal and rheological behavior of a glycerol-water plasticized starch formulation. The goal was to determine how levels of edible film forming components impacted gelatinization temperature, time, and viscosity. In Trial 1 1.8g of starch was used along with 12g of glycerol and 46.2g of water. These inputs yielded a gelatinization temperature of 65 degrees centigrade that was attained in 5 minutes 9 seconds with a viscosity of 120.4cp. This was set as the baseline with the lowest level of 1.8g starch.

An improvement for Trial 2 was set at 15g of glycerol and 42.6g of water with 2.4g of starch. The level measured in Trial 1 was set at 2.4g. The results indicate value to be 68 degrees centigrade in temperature, 5 minutes 56 seconds in time and 148.7 cp for viscosity,

supporting greater starch interactions. Trial 3 used 3g of starch with 18g of glycerol and 39g of water which is given in the Table 4.2. The results reached 70 degrees centigrade with 6 minutes and 51 seconds in time and a viscosity of 174.2 cp. This increase is suggested by greater swelling of starch granules. The data showed that greater levels of starch led to an increase in swelling of granules and molecular interaction leading to increased gelation temperature and time. Glycerol however improved the flexibility of the material.

4.3 RESPONSE SURFACE METHODOLOGY

Table 4.3Responses for RSM Trials for the developed edible film

Run	Component 1 Strach (g)	Component 2 Glycerol (g)	Component 3 Water (g)	Response 1 Gelatinization Temperature ℃	Response 2 Gelatinization Time min	Response 3 Viscosity cp
1	2.23442	17.7656	40	66	5.8	140.2
2	3	15.01	41.99	68	6.2	155.3
3	1.955	12.9255	45.1195	65	5.1	130.1
4	2.23442	17.7656	40	66	5.8	140.7
5	1.8	13.8933	44.3067	64	4.9	129.5
6	3	13.0162	43.9838	64	4.8	132.8
7	2.45479	14.2981	43.2471	67	6.2	145.3
8	3	12	45	69	6.5	159.7
9	1.8	15.9426	42.2574	65	5.3	135.2
10	3	16.4138	40.5862	66	5.4	137.6
11	1.8	17.0425	41.1575	65	5.5	140.4
12	1.955	12	46.045	64	5.1	128.7
13	2.45479	14.2981	43.2471	67	6	145.3
14	3	16.4138	40.5862	65	5.4	137.9
15	2.45479	14.2981	43.2471	67	6.1	148.2
16	2.45479	14.2981	43.2471	67	6.2	146.8

The experimental design was on a blend of three components: starch, glycerol, and water. The specific component constraints were as follows: starch was between 1.8 to 3 grams, glycerol was between 12 to 18 grams, and water was between 40 to 46.2 grams, with each of the three formulations weighing a total of 60 grams. These were imposed to systematically study their influence on functional responses like gelatinization temperature, gelatinization time, and viscosity. 16 experimental runs were performed, adjusting the proportion of each component in the specified range. Data table obtained reflected corresponding changes in response parameters. The gelatinization temperature varied between 64 and 69°C, gelatinization time from 4.8 to 6.5 minutes, and viscosity from 128.7 to 159.7 cp. All these variations revealed the dominance of formulation composition on the physical and functional characteristics of the starch-based system. This data provided the foundation for surface response analysis and statistical modeling.

4.3.1 Effectson these compounds by the Gelatinization Temperature

Table 4.4ANOVA for linear model (Response 1: Gelatinization temperature)

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	31.58	8	3.95	20.37	0.0003	significant
[(1)]Linear Mixture	7.99	2	4.00	20.62	0.0012	
AB	2.76	1	2.76	14.25	0.0069	
AC	3.75	1	3.75	19.37	0.0032	
BC	0.7257	1	0.7257	3.74	0.0942	
A^2BC	0.5588	1	0.5588	2.88	0.1333	
AB^2C	1.00	1	1.00	5.17	0.0572	
ABC^2	3.36	1	3.36	17.34	0.0042	
Residual	1.36	7	0.1938			
Lack of Fit	0.8568	2	0.4284	4.28	0.0824	not significant
Pure Error	0.5000	5	0.1000			
Cor Total	32.94	15				

Table 4.5Fit statistics of the (Response 1: Gelatinization temperature)

Std. Dev.	0.4403	**R^2**	0.9588
Mean	65.94	**Adjusted R^2**	0.9117
C.V. %	0.6677	**Predicted R^2**	-6.4372
		Adeq Precision	14.7761

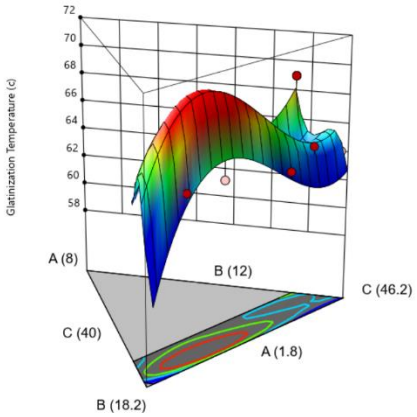


Fig 4.1Effects on these compounds by the Gelatinization Temperature

The ANOVA for the special quartic model testing gelatinization temperature has statistical significance with an F-value of 20.37 and a p-value of 0.0003, reflecting efficient explanation of variation due to component interactions. The Linear Mixture term is the largest contributor (F = 20.62, p = 0.0012), reflecting a very strong individual effect. Important interactions are AB (F = 14.25, p = 0.0069), AC (F = 19.37, p = 0.0032), and the second-order term ABC^2 (F = 17.34, p = 0.0042), reflecting combined influences on gelatinization temperature. BC (p = 0.0942) and A^2BC (p = 0.1333) were not significant. The Lack of Fit test was not significant (p = 0.0824), indicating model adequacy. Good fit is reflected through high R^2 (0.9588) and adjusted R^2 (0.9117), but negative predicted R^2 (-6.4372) indicates possible overfitting which is given the Table (4.4 and 4.5). However, Adequate Precision 14.7761 reflects an excellent signal-to-noise ratio, confirming model reliability.

4.3.2 Effects on these compounds by the Gelatinization Time

Table 4.6 ANOVA for linear model (Response 2: gelatinization time)

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	4.12	8	0.5154	100.82	< 0.0001	significant
(1)Linear Mixture	0.5613	2	0.2807	54.90	< 0.0001	
AB	0.2372	1	0.2372	46.39	0.0003	
AC	0.3484	1	0.3484	68.14	< 0.0001	
BC	0.0113	1	0.0113	2.22	0.1798	
A^2BC	0.0113	1	0.0113	2.21	0.1809	
AB^2C	0.2930	1	0.2930	57.31	0.0001	
ABC^2	0.2352	1	0.2352	46.01	0.0003	
Residual	0.0358	7	0.0051			
Lack of Fit	0.0083	2	0.0041	0.7533	0.5177	not significant
Pure Error	0.0275	5	0.0055			
Cor Total	4.16	15				

Table 4.7 Fit statistics of the (Response 2: gelatinization time)

Std.Dev	0.0715	**R^2**	0.9914
Mean	5.64	**Adjusted R^2**	0.9816
C.V %	1.27	**Predicted R^2**	0.2270
		Adeq Precision	31.4485

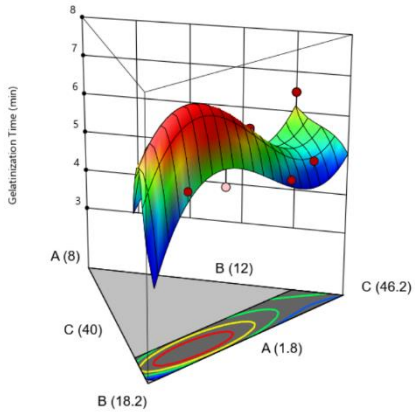


Fig 4.2Effects on these compounds by the Gelatinization Time

ANOVA results for the special quartic model of gelatinization time are significant with an F-value of 100.82 and $p < 0.0001$, validating the significant impact of the model terms. The Linear Mixture component is significant ($F = 54.90$, $p < 0.0001$), indicating the significant impact of base composition. Important interaction terms are AB ($F = 46.39$, $p = 0.0003$), AC ($F = 68.14$, $p < 0.0001$), AB^2C ($F = 57.31$, $p < 0.0001$), and ABC^2 ($F = 46.01$, $p = 0.0003$), all significantly influencing gelatinization time. BC ($p = 0.1798$) and A^2BC ($p = 0.1809$) were not significant. The Lack of Fit test ($p = 0.5177$) indicates a good model fit. With an R^2 of 0.9914 and Adjusted R^2 of 0.9816, the model demonstrates high correlation. Although the Predicted R^2 (0.2270) is low, it still indicates some predictability. Which is given in the Table (4.6 and 4.7). The Adequate Precision value of 31.4485 validates a high signal-to-noise ratio and model strength. Generally, the model accurately represents important factors but could be improved with further refinement for better prediction.

4.3.3 Effects on these compounds by the Viscosity

Table 4.8 ANOVA for linear model (Response 3: Viscosity)

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	1186.03	8	148.25	65.82	< 0.0001	significant
[(1)]Linear Mixture	336.62	2	168.31	74.72	< 0.0001	
AB	40.92	1	40.92	18.17	0.0037	
AC	69.85	1	69.85	31.01	0.0008	
BC	4.91	1	4.91	2.18	0.1833	
A^2BC	9.68	1	9.68	4.30	0.0768	
AB^2C	67.74	1	67.74	30.07	0.0009	
ABC^2	75.44	1	75.44	33.49	0.0007	
Residual	15.77	7	2.25			
Lack of Fit	9.78	2	4.89	4.08	0.0889	not significant
Pure Error	5.99	5	1.20			
Cor Total	1201.80	15				

Table 4.9 Fit statistics of the (Response 3: Viscosity)

Std.Dev	1.50	**R^2**	0.9869
Mean	140.86	**Adjusted R^2**	0.9719
C.V %	1.07	**Predicted R^2**	-2.0455
		Adeq Precision	27.5666

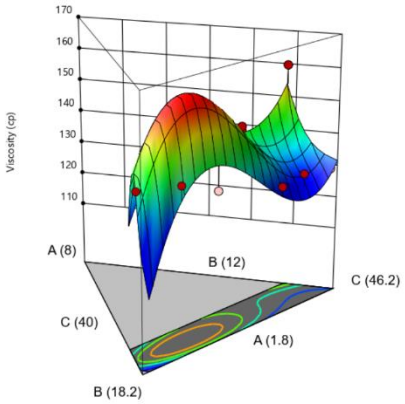


Fig 4.3Effects of Viscosity

The ANOVA outcome for the special quartic model for evaluating viscosity is highly statistically significant, with F = 61.36 and $p < 0.0001$, reflecting the strong impact of the independent variables on viscosity. The Linear Mixture term is also significantly high (F = 66.35, $p < 0.0001$). The important interactions are AB (F = 18.08, p = 0.0038), AC (F = 30.53, p = 0.0009), AB^2C (F = 27.19, p = 0.0012), and ABC^2 (F = 33.30, p = 0.0007), all adding significantly to viscosity. BC and A^2BC were not significant ($p > 0.05$), indicating negligible effect. The Lack of Fit test (p = 0.0689) validates that the model is a good fit for the data. The model has excellent explanatory power with an R^2 of 0.9859 and Adjusted R^2 of 0.9699. Although a negative Predicted R^2 (-2.5764) suggests limited predictive capability, the Adequate Precision value of 26.7096 indicates a strong signal and model stability which is

given in the Table (4.8 and 4.9). The model is able to capture the effects on viscosity well, although additional refinements might further improve predictive accuracy.

Table 4.10Solution obtained from the Response surface methodology

Number	Starch g	Glycerol g	Water g	Gelatinization Temperature ℃	Gelatinization Time min	Viscosity cp	Desirability	
1	**2.994**	**12.000**	**45.006**	**69.001**	**6.509**	**159.638**	**0.998**	**Selected**
2	2.448	12.042	45.510	69.581	6.700	156.721	0.785	
3	2.397	12.000	45.603	69.563	6.700	156.441	0.763	
4	2.958	15.056	41.986	68.195	6.332	156.363	0.750	

4.4 RESULT COMPOSITON OF THE OPTIMIZED EDIBLE FILM

Table 4.11 Composition of ingredients for the developed edible film

Materials	Optimized composition
Starch	3g
Glycerol	12g
Water	45g



Fig 4.4 Developed edible film

4.5ASH CONTENT

Edible film of tapioca starch, glycerol, and water had ash content of 2.196% and was within the normal range (1.5 to 2.5) for starch-derived films. It indicates acceptable quality and mineral level, with elevated value likely coming from trace minerals in the tapioca starch or impurities on film formation on a very minute scale. As glycerol is a chemically pure organic compound, there is no ash contribution. Ash content shows low inorganic matter presence, which is favorable for food packaging since it reduces effects on flavor, color, and nutrition. At 2.196% ash content, the film formulation and processing are effective, and the outcome is within industry requirements for edible and food-contact purposes.

4.6 CARBOHYDRATES

Carbohydrate content in the starch which is isolated employing the sedimentation process is 88%, owing to the major carbohydrate composition in tapioca starch, mainly amylose and amylopectin. It strengthens the mechanical integrity of the film, film forming,

and barriers. Glycerol, as a sugar alcohol, does not contribute to it. This carbohydrate content is consistent with comparable starch-based films (60-80%) and reflects that the film is nutritionally dense in carbohydrates, which is advantageous for food-contact uses. Carbohydrates also influence flexibility, solubility, and moisture permeability, which are significant for packaging functionality. Overall, the 88% carbohydrate content verifies the starch-based composition and validates the film's promise as a biodegradable, edible packaging material.

4.7WATER ACTIVITY

The water activity (a_w) of the edible film produced from tapioca starch, glycerol, and water was 0.621a_w, which is a moderate level of free water. This has implications in microbial stability, chemical reactivity, and shelf life. The aw value of above 0.7 could encourage growth of yeast and mold if improperly stored. The relatively high a_w is attributed to glycerol, a hygroscopic plasticizer that absorbs water and lowers the water-binding capacity of starch. The effect is to enhance flexibility but enhance free water availability. The 12% moisture content of the film confirms this, yielding a soft but microbially vulnerable film. This level of water activity increases elasticity but can decrease barrier properties such as water vapor resistance. These films are more appropriate for storage in low-humidity or low-moisture food applications. Although higher than the usual range of 0.3–0.6 a_w for dehydrated edible films, 0.621a_w can be tolerated based on use. Compromises such as lowering glycerol, longer drying, or the inclusion of antimicrobials might enhance shelf stability.

4.8TENSILE STRENGTH

The tensile strength of the film from tapioca starch, glycerol, and water was determined as 826.5 N/m, which is within moderate mechanical strength levels common in glycerol plasticized starch-based films. The figure obtained signifies the resistance of the film to breakage under tension, which is ideal for manipulation and packaging under minimal pressure. Glycerol, used as a plasticizer, adds flexibility by breaking hydrogen bonding among starch chains but can decrease tensile strength. Yet, the ratio of glycerol to starch in this formulation preserves a strong matrix. In general, the tensile strength of 826.5 N/m supports the film's application in low- to moderate-moisture food systems. Future refinements might include varying glycerol levels or drying conditions to maximize strength for particular commercial applications.

4.9TEAR RESISITANCE

The tear resistance of the tapioca starch-based edible film containing glycerol and water was 7.875 mN, reflecting a degree of resistance to tear propagation. This moderate level indicates that the film might be more vulnerable to tearing on stress or rough treatment. Tear behavior depends on the starch–glycerol ratio. Although glycerol improves flexibility, it decreases structural integrity so that the film becomes more likely to tear if highly plasticized. Based on the measured resistance, the film is appropriate for low-stress applications but can be optimized, e.g., by decreasing glycerol levels, reducing moisture, or modifying drying conditions, to enhance tear strength for general packaging applications.

4.10WATER VAPOUR PERMEABILITY OF EDIBLE FILM

The water vapor permeability (WVP) of the tapioca starch-based edible film developed using glycerol and water was found to be 459.183 g/m²/day, reflecting moderate permeability. This implies the film facilitates observable moisture transfer but is less efficient than synthetic options optimized for high moisture resistance. The higher WVP affects the film's 12% moisture content, along with glycerol, as it disrupts the interactions of starch and creates free volume, which enhances water vapor diffusion. Moreover, the hydrophilic character of starch is also responsible for higher permeability. Although the WVP is greater than that of synthetic films, it is within the range to be expected for starch-based edible films, particularly those with tapioca starch and glycerol. The films are good for biodegradability and flexibility but might not be best suited for long-term storage or high-moisture conditions. In conclusion, the WVP is in favor of the film's application in short-term or low-humidity uses. For improved barrier characteristics, future developments may involve decreasing glycerol, decreasing moisture, or incorporating hydrophobic additives.

4.11WATER ABSORPTION CAPACITY OF EDIBLE FILM

The moisture absorption of the edible film from tapioca starch, glycerol, and water was found to be 129 g/m², which is a moderate to high level of moisture absorption. This is an indication of the hydrophilic character of the film, which describes its tendency when exposed to humid or wet conditions. The reason for the high absorption is primarily the starch, which has hydrogen bonds with water molecules, and glycerol, which is hygroscopic and increases moisture absorption. To enhance moisture resistance, subsequent treatments may lower starch content, maximize glycerol concentration, or incorporate hydrophobic agents. Generally, the water uptake profile aids in its functional benefits and reveals areas to be optimized.

4.12 BIODEGRADABILITY OF THE EDIBLE FILM

The film from tapioca starch, glycerol, and water showed noticeable texture and brittleness changes over time. The film, fig(4.5) after about 20 days, turned brittle and lost its flexibility and acceptable texture. This would be expected as a result of moisture loss and the breakdown of the plasticized starch matrix. To determine biodegradability, the film was buried in the ground for approximately three weeks. The film exhibited clear signs of decomposition during this time, which indicates its biodegradability under natural soil conditions. Structural integrity loss and visible disintegration are evidence that the film is eco-friendly and can be used where biodegradability is necessary.



Fig 4.5Biodegradability of the edible film

CHAPTER5

CONCLUSION

The study showed the potential of tapioca starch as a renewable raw material for biodegradable, edible films. Of the three extraction processes—acid, alkali, and sedimentation—sedimentation was most effective for collecting film-forming starch. The concentrations of starch and glycerol were optimized using Response Surface Methodology (RSM) in order to optimize film properties. The movies showed good physicochemical and mechanical properties such as good gelatinization temperature and time, moderate viscosity, good tensile and tear resistance, low water vapor permeability, and good water absorption. The low ash and carbohydrate content showed the purity of the starch, while water activity and ISO-approved testing ensured the quality of the film. The optimized film demonstrated excellent biodegradability and structural integrity, offering a viable alternative to plastic packaging. These findings validate the application of starch-based films in minimizing plastic pollution and emphasize their potential in sustainable food packaging.

REFERENCES

1. Aghazadeh M., Karim R., Rahman R. A., Sultan M. T., Johnson S. K., and Paykary M. (2018) 'Effect of glycerol on the physicochemical properties of cereal starch films', Czech Journal of Food Sciences, Vol. 36 No.5.

2. Basiak E., Lenart A., and Debeaufort F. (2018) 'How glycerol and water contents affect the structural and functional properties of starch-based edible films', Polymers, Vol. 10 No.4, pp. 412.

3. Bergo P., Sobral P. J. D. A., and Prison J. M. (2008) 'Effect of glycerol on physical properties of cassava starch films', Journal of Food Processing and Preservation, Vol. 34, pp. 401–410.

4. Chiusarelli M., and Hubinger M. D. (2014) 'Evaluation of edible films and coatings formulated with cassava starch, glycerol, carnauba wax, and stearic acid', Food Hydrocolloids, Vol. 38, pp. 20–27.

5. Chillo S., Ventrella A., Bevilacqua A., and Gabriele D. (2008) 'Influence of glycerol and chitosan addition on the properties of tapioca starch-based edible films', Carbohydrate Polymers, Vol. 74 No.2, pp. 287–295.

6. Cisse M., Fall M., Diop A., and Ndiaye M. (2012) 'Effect of varying concentrations of chitosan and glycerol on the properties of edible films', Food Research International, Vol. 47 No.1, pp. 319–327.

7. Farahnaky A., Saberi B., and Majzoobi M. (2013) 'Effect of glycerol on physical and mechanical properties of wheat starch edible films', Journal of Texture Studies, Vol. 44 No.3, pp. 176–186.

8. Fama M., Dey M. S., and Dutta J. (2005) 'Mechanical properties of tapioca starch-based edible films containing sorbates', Food Science and Technology International, Vol. 11 No.4, pp. 249–254.

9. Gutierrez M., Salgado P. R., and Figueroa F. L. (2015) 'Structural and mechanical properties of edible films produced from native and modified starches of cush-cush yam and cassava', Carbohydrate Polymers, Vol. 121, pp. 224–233.

10. Juliyarsi S., Setiawati R., and Atmaja M. (2011) 'Evaluation of the quality of edible films using glycerol as a plasticizer', International Journal of Food Science and Technology, Vol. 46 No.5, pp. 1195–1202.

11. Kaveh M., Harvima I. T., and Sari K. (2020) 'Effects of alcoholic-alkaline treatments on the physical and mucoadhesive properties of tapioca starch', Starch- Vol. 72 No.1, pp. 53–59.

12. Kumar A. (2019) 'Review of polysaccharide-based films for food packaging', Journal of Food Engineering, Vol. 271, pp. 1–9.

13. Kusumaningtyas R. D., Putri R. D., Badriah N., and Faizah F. E. (2018) 'Preparation and characterization of edible film from sorghum starch with glycerol and sorbitol as plasticizers', Journal of Engineering Science and Technology, Vol. 13, pp. 47–55.

14. Maran J. P., Gopinath S. C. B., and Raj S. (2013) 'Mathematical modeling of mechanical properties of edible films based on tapioca starch', Journal of Food Engineering, Vol. 116 No.2, pp. 349–357.

15. Parra J., Moreira R. G., and Solis S. (2004) 'Mechanical behavior and water vapor transmission rates of cassava starch-based edible film blends', Carbohydrate Polymers, Vol. 57 No.4, pp. 459–466.

16. Pansaeng R. (2019) 'Effect of glycerol concentration on the physical and mechanical properties of biodegradable films made from cassava starch (Saai Dieow cultivar)', Journal of Agricultural and Food Chemistry, Vol. 67 No.18, pp. 5124–5130.

17. Piyawatakarn S., Jong P., and Srichana T. (2015) 'Effect of glycerol on properties of tapioca starch-based films', Food Science and Technology, Vol. 45 No.2, pp. 305–312.

18. Prabha T., and Ranganathan P. (2018) 'Optimization of barrier properties of biodegradable films made from tapioca starch', Food Packaging and Shelf Life, Vol. 17, pp. 20–29.

19. Rumyaan M., Polnaya F. J., and Tuhumury H. C. (2022) 'Characteristics of edible film made from bitter cassava starch with glycerol', AGRITEKNO: Journal TechnologyPertanian, Vol. 11 No.2, pp. 102–107.

20. Saberi B., Farahnaky A., and Karami Z. (2016) 'Mechanical and physical properties of pea starch edible films with varying concentrations of glycerol', International Journal of Food Science and Technology, Vol. 51 No.6, pp. 1412–1418.

21. Sartika R. S., Gita D., and Yuliana M. (2022) 'Review on tensile properties of edible films made from various starches and glycerol', Journal of Food Engineering, Vol. 214, pp. 87–95.

22. Wilfer J., Bücking M., and Rummel M. (2021) ‘Effect of native starch types on the properties of edible films’, Food Biophysics, Vol. 16 No.4, pp. 321–328.

23. Zhong Y., and Li Y. (2014) ‘Effects of glycerol and storage relative humidity on the properties of kudzu starch-based edible films’, Starch, Vol. 66 No.5–6, pp. 524–532.